

Report
Of
Gen AI Virtual Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

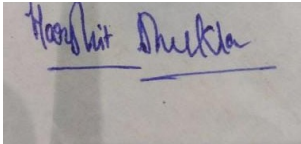
Roll Number of Student: 2410031071

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, HARSHIT SHUKLA, Roll No. 2410031071, do hereby declare that the internship report titled “Gen AI Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.



Signature

Student Name: Harshit Shukla

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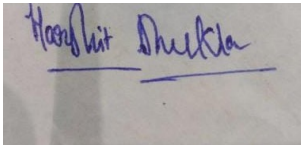
Batch: 2024-2028

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Gen AI Virtual Internship Program organized by AICTE, EduSkills and the National Internship Portal, with curriculum provided by AWS Academy. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would also like to thank my faculty mentors and the Department at IILM University, Greater Noida, for their continuous guidance, support, and encouragement throughout the duration of this internship.

I express my gratitude to my parents for their blessings, and to my family and peers for their constant motivation and support.



Signature

Student Name: Harshit Shukla

Roll No.: 2410031071

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that the Gen AI Virtual Internship Report submitted by Harshit Shukla (Roll No. 2410031071), a student of B.Tech (CSE), Batch 2024-2028, IILM University, Greater Noida, is a record of the 8-week virtual internship titled “Gen AI Virtual Internship” undertaken during June–August 2026, organized under the National Internship Portal by AICTE and EduSkills, with curriculum provided by AWS Academy. The internship was successfully completed and graded 'O' (Outstanding), as evidenced by the certificate reproduced below (Certificate ID: 4bdc582a40842610707).



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4. PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence has rapidly transformed from a niche academic discipline into one of the most influential technologies shaping industries worldwide. Among its many branches, Generative AI (Gen AI) has emerged as a particularly transformative field, enabling machines to create text, images, code, and other content that closely resembles human-generated output. Recognizing the growing demand for industry-ready skills in this domain, the All India Council for Technical Education (AICTE) and EduSkills, under the aegis of the Ministry of Education, launched the Gen AI Virtual Internship programme on the National Internship Portal, with technical curriculum delivered by AWS Academy.

This 8-week virtual internship was designed to bridge the gap between academic learning and industry requirements by exposing students to real-world applications of Generative AI built on Amazon Web Services (AWS). As a student of B.Tech CSE, Batch 2024-2028, at IILM University, Greater Noida, I undertook this internship during June–August 2026 to strengthen my understanding of cloud computing and Generative AI, and to gain hands-on exposure to industry-standard tools and platforms.

4.2 Organization Profile

The Gen AI Virtual Internship was organized jointly by the following bodies:

- AICTE (All India Council for Technical Education) – The apex statutory body for technical education in India, under the Ministry of Education.
- National Internship Portal – A government platform connecting students with industry-relevant internship opportunities.
- EduSkills – A skilling foundation working with AICTE to bring corporate-designed curricula to engineering and management students, under the initiative “Nation Building Through Skills.”
- AWS Academy – Amazon Web Services' global education programme that provides free, ready-to-teach cloud computing and AI curriculum to academic institutions.

The Gen AI Virtual Internship itself is a structured, self-paced online programme that combines conceptual learning with practical, lab-based exercises, delivered entirely through the AWS Academy learning platform.

4.3 Problem Statement

With Generative AI rapidly becoming central to modern software and cloud ecosystems, there exists a significant gap between theoretical classroom knowledge and the practical, industry-relevant skills required to design, build, and deploy Generative AI applications. This internship addressed the problem of equipping a 3rd-year B.Tech student with hands-on exposure to cloud computing fundamentals, machine learning concepts, and Generative AI tools on the AWS platform, so as to become industry-ready.

4.4 Project Objectives

- To develop a strong conceptual foundation in Artificial Intelligence, Machine Learning, and Generative AI.
- To gain hands-on experience with AWS cloud services relevant to building and deploying Generative AI applications.
- To understand the architecture and working of Large Language Models (LLMs) and foundation models.
- To learn practical prompt engineering techniques for effectively interacting with generative models.
- To explore real-world use cases of Generative AI across industries such as healthcare, finance, retail, and education.
- To apply theoretical knowledge through guided labs and hands-on exercises on the AWS platform.
- To enhance employability by acquiring an industry-recognized skill set and certification.

4.5 Scope of the Project

The scope of this internship encompassed an 8-week structured curriculum covering cloud computing fundamentals, machine learning and deep learning concepts, and applied Generative AI development using AWS services such as Amazon Bedrock and

Amazon SageMaker. It focused on building conceptual understanding as well as practical, lab-based skills that are applicable across a wide range of industries, without extending into production-scale deployment or enterprise-level system integration.

4.6 Technologies and Tools Used

- Amazon Web Services (AWS) core services: EC2, S3, IAM, VPC.
- Amazon SageMaker – for AI/ML model building and workloads.
- Amazon Bedrock – for accessing foundation models and Large Language Models (LLMs) via API.
- AWS Academy Learning Platform – for curriculum delivery and guided labs.
- Concepts of tokenization, embeddings, transformers, and prompt engineering.

4.7 System Architecture

The Generative AI applications explored during the internship followed a typical cloud-native architecture: a user-facing interface sends prompts to a foundation model hosted and served through Amazon Bedrock; the foundation model, built on transformer architecture, processes the tokenized input and generates output such as text, summaries, or conversational responses. Supporting AWS services such as S3 (for data storage), IAM (for access control), and SageMaker (for custom model training and evaluation) worked alongside Bedrock to provide a complete, scalable pipeline for building and testing Generative AI use cases.

4.8 Methodology

The internship curriculum was structured progressively over 8 weeks, moving from foundational concepts to applied Generative AI development:

- Week 1–2: Foundations of Cloud Computing and AI – introduction to cloud computing concepts, AWS global infrastructure, core AWS services (EC2, S3, IAM, VPC), and AI/ML fundamentals.
- Week 3–4: Machine Learning and Deep Learning Concepts – supervised, unsupervised, and reinforcement learning paradigms, neural networks, and an overview of Amazon SageMaker.

- Week 5–6: Introduction to Generative AI and Foundation Models – foundation models, transformers, LLMs, Amazon Bedrock, tokenization, embeddings, and model fine-tuning.
- Week 7: Prompt Engineering and Responsible AI – zero-shot, few-shot, and chain-of-thought prompting techniques, along with fairness, transparency, bias mitigation, and data privacy principles.
- Week 8: Applied Generative AI and Capstone Activity – hands-on labs building Generative AI use cases such as text generation, summarization, and chatbot-style interactions, followed by review, assessment, and certification.

4.9 Expected Outcomes

Upon completion of the internship, the following outcomes were achieved:

- A working knowledge of cloud computing fundamentals and core AWS services (EC2, S3, IAM).
- Conceptual clarity on Machine Learning, Deep Learning, and Generative AI, including LLMs and foundation models.
- Practical exposure to Amazon Bedrock and Amazon SageMaker for building AI/ML and Generative AI solutions.
- Proficiency in prompt engineering techniques for effective interaction with generative models.
- Improved self-directed learning, time management, analytical thinking, and technical documentation skills.

Successfully completing this internship with an 'Outstanding' grade strengthened confidence in the domain of Artificial Intelligence and cloud computing, and has motivated further learning and projects in Generative AI, supporting both academic growth and long-term career aspirations in the technology sector.

4.10 Certificate of Completion and Other Communication Mail Proof

The certificate of completion issued by AICTE, EduSkills, and AWS Academy for this internship is attached below as proof of successful completion.



5. BIBLIOGRAPHY / REFERENCES

1. AICTE – National Internship Portal, Ministry of Education, Government of India.
2. EduSkills Foundation – “Nation Building Through Skills” Initiative.
3. AWS Academy – Amazon Web Services Educational Curriculum.
4. Gen AI Virtual Internship – Course Modules and Lab Guides (AWS Academy, 2026).